

Notes on RoPE

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27/05/2015

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The Concept of Positional Encoding

Definition

- Positional encoding captures the sequential order information of words in NLP
- It incorporates position information during the transformation from a token embedding to its query, key, and value for attention computation

Models in NLP and their positional encoding

- RNN: Computing the feature vector of an input (word) with the intermediate output of the previous one
- CNN: Position-agnostic
- PLM (Su et al. 2023) (pre-trained language model): Position-agnostic

The Modeling of Positional Encoding

$$\mathbf{y}_i^t = \begin{cases} W_t \cdot \mathbf{x}_i & \text{without positional encoding} \\ f_t(\mathbf{x}_i, i) & \text{with positional encoding,} \end{cases} \quad (1)$$

where $W_t \in \mathbb{R}^{d_{model} \times d_{model}}$, $\mathbf{x}_i \in \mathbb{R}^{d_{model}}$ is the embedding of a token in a certain layer, $t \in \{Q, K, V\}$, $i \in \{1, 2, \dots, L\}$, and L is the maximum sequence length.

The Modeling of Positional Encoding

- The computation of attention involves the inner product of query and key at all possible combinations of position pairs in a sequence, depending on a pre-defined mask.
- Therefore, the position indexes of the query and the key (and value) are denoted m and n respectively

$$\mathbf{q}_m = \mathbf{y}_i^q = f_q(\mathbf{x}_m, m) \quad (2)$$

$$\mathbf{k}_n = \mathbf{y}_i^k = f_k(\mathbf{x}_n, n) \quad (3)$$

$$\mathbf{v}_n = \mathbf{y}_i^v = f_v(\mathbf{x}_n, n), \quad (4)$$

where $m, n \in \{1, 2, \dots, L\}$.

Existing Positional Encoding for PLM

- Absolute position encoding
 - Generated through a pre-defined function
 - Trainable encoding
- Relative position encoding
- Modeling based on Neural ODE
- Modeling in complex space

The common aspects of all the above methods

- Adding the position encoding to the input embedding
- Not suitable for linear self-attention

Absolute Positional Encoding

- The transformation of a token embedding

$$f_t(\mathbf{x}_i, i) = W_t \cdot (\mathbf{x}_i + \mathbf{p}_i), \quad (5)$$

where $\mathbf{p}_i \in \mathbb{R}^{d_{model} \times 1}$ is the embedding of position information.

- With a pre-defined function (Vaswani et al. 2023)

$$\mathbf{p}_{i,k} = \begin{cases} \sin(i/10000^{2j/d_{model}}), & k = 2j \\ \cos(i/10000^{2j/d_{model}}), & k = 2j + 1, \end{cases} \quad (6)$$

where $k \in \{0, 1, \dots, d_{model} - 1\}$.

Remarks

- It is the same set of vectors, i.e. $\{\mathbf{p}_i | i = 1, 2, \dots, L\}$, for Q, K, and V.
- Note that between $(k = 2j \text{ or } 2j + 1)$ and $(k = 2j \text{ or } 2j - 1)$, as long as the training and inference uses the same formulation, both work.
- The former is used here to make the dimension index k zero-based, consist with that in (Vaswani et al. 2023; Su et al. 2023)

Relative Positional Encoding

- Learned positional encoding to K and V

$$f_t(\mathbf{x}_i, i) = W_t \cdot (\mathbf{x}_i + \tilde{\mathbf{p}}_s^t), \quad (7)$$

where $\tilde{\mathbf{p}}_s^t$ are learned encoding vectors, $t \in \{K, V\}$, and s is the relative distance between a query vector $\mathbf{q}_m$ and a key vector $\mathbf{k}_n$, i.e.

$$s = \text{clip}(m - n, s_{\min}, s_{\max}). \quad (8)$$

How s is used during the computation of attention?

- $\{\tilde{\mathbf{p}}_s^Q\}, \{\tilde{\mathbf{p}}_s^K\}$ are learned for all possible relative distances $\{s | s_{\min} \leq s \leq s_{\max}, s \in \mathbb{Z}_{\geq 0}\}$
- When computing an attention weight between $\mathbf{q}_m$ and $\mathbf{k}_n$, which is $f_Q(\mathbf{x}_m)^\top f_K(\mathbf{x}_n, n)$, $f_K(\mathbf{x}_n, n) = W_K \cdot (\mathbf{x}_n + \tilde{\mathbf{p}}_{(m-n)}^K)$, $f_Q(\mathbf{x}_m) = W_Q \mathbf{x}_m$
- Encoding directly in attention weight
- RoPE

Relative Positional Encoding Directly in Attention Weight

- Based on (2), (3), and (5),

$$\begin{aligned}\mathbf{q}_m^\top \cdot \mathbf{k}_n &= (W_Q(\mathbf{x}_m + \mathbf{p}_m))^\top (W_K(\mathbf{x}_n + \mathbf{p}_n)) \\ &= (\mathbf{x}_m + \mathbf{p}_m)^\top (W_Q^\top W_K)(\mathbf{x}_n + \mathbf{p}_n) \\ &= \mathbf{x}_m^\top W_Q^\top W_K \mathbf{x}_n + \mathbf{x}_m^\top W_Q^\top W_K \mathbf{p}_n + \mathbf{p}_m^\top W_Q^\top W_K \mathbf{x}_n + \\ &\quad \mathbf{p}_m^\top W_Q^\top W_K \mathbf{p}_n\end{aligned}\tag{9}$$

- Replacing the absolute position encoding with relative position encoding

$$\begin{aligned}\mathbf{q}_m^\top \cdot \mathbf{k}_n &= \mathbf{x}_m^\top W_Q^\top W_K \mathbf{x}_n + \mathbf{x}_m^\top W_Q^\top \widetilde{W}_K \widetilde{\mathbf{p}}_{m-n} + \mathbf{u}^\top W_Q^\top W_K \mathbf{x}_n + \\ &\quad \mathbf{v}^\top W_Q^\top \widetilde{W}_K \widetilde{\mathbf{p}}_{m-n}\end{aligned}\tag{10}$$

Rotary Position Embedding (RoPE)

Desirable properties of positional encoding (need more details)

- The sequence length flexibility
- Decaying inter-token dependency with increasing relative distances
- The capability of equipping the linear self-attention with relative position encoding

RoPE

- Encoding the absolute position for each token with a rotation matrix
- The attention weight only relies on the relative position dependency
- The encoding is applied to multiple transformer layers
- Leveraging the positional information into the learning process of PLM
- Distinguishing itself due to the above desirable properties

Derivation of RoPE in $\mathbb{R}^2$

- The goal is to find the transformation of the token embedding $f_q(\mathbf{x}_q, m)$ and $f_k(\mathbf{x}_k, n)$ for the query and key, between which **the inner product, i.e. the attention weight, depends only the input embedding $\mathbf{x}_q$ and $\mathbf{x}_k$, and the relative position $m - n$** , where $m, n \in \{1, 2, \dots, L\} \cup \{0\}$ (refer to next page for more details), i.e.

$$\mathbf{q}_m^\top \mathbf{k}_n = \langle f_q(\mathbf{x}_q, m), f_k(\mathbf{x}_k, n) \rangle = g(\mathbf{x}_q, \mathbf{x}_k, n - m). \quad (11)$$

The change from $f_t(\mathbf{x}_m, m)$ in (2,3) to $f_t(\mathbf{x}_t, m)$ in (11)

When $m = 0$, a formulation of the transformation of the token embedding without position information in any position $m \in \{1, 2, \dots, L\}$ is needed, yet $f_t(\mathbf{x}_m, m)|_{m=0} = f_t(\mathbf{x}_0, 0)$, where $t \in \{Q, K\}$, can only represent the transformation in the position $m = 0$.

The meaning of $= g(\mathbf{x}_m, \mathbf{x}_n, n - m)$

It only specifies the conditions the inner product should meet as stated above, not a specific parametric form of it.

Derivation of RoPE in $\mathbb{R}^2$

- $m \neq 0, n \neq 0$: Other than the input token embedding, the inner product depends only on the position difference
 - $m \neq n$: Inner product between the **position encoded** transformation of $\mathbf{x}_q$ and $\mathbf{x}_k$ in **different** positions $m, n \in \{1, 2, \dots, L\}$
 - $m = n$: Inner product between the **position encoded** transformation of $\mathbf{x}_q$ and $\mathbf{x}_k$ in the **same** position $m \in \{1, 2, \dots, L\}$
- $m = 0, n = 0$: Other than the input token embedding, the inner product has no other dependency, because the transformation itself is computed without position information at all
 - Inner product between the transformation **without position information** of $\mathbf{x}_q$ and $\mathbf{x}_k$ in any position $m, n \in \{1, 2, \dots, L\}$.
 $m = n = 0$ does not refer to the position of the 0 – *th* token as the position starts from 1. It simply indicates that the transformation does not contain any information on position.
- $(m \neq 0, n = 0)$ and $(m = 0, n \neq 0)$: Invalid combinations because there is no attention weight by the inner product between a position encoded transformation and one without position information.

Derivation of RoPE in $\mathbb{R}^2$

- For the first case in (Pg. 12), the **position encoded** transformation of the token embedding for the query in position m and that for the key in position n , represented by $f_q(\mathbf{x}_q, m)$ and $f_k(\mathbf{x}_k, n)$ in (11), where $m, n \in \{1, 2, \dots, L\}$, can be formulated using complex form as follows

$$\mathbf{q}_m = \|\mathbf{q}_m\| e^{i\theta_{q_m}} = f_q(\mathbf{x}_q, m) = R_q(\mathbf{x}_q, m) e^{i\Theta_q(\mathbf{x}_q, m)} \quad (12)$$

$$\mathbf{k}_n = \|\mathbf{k}_n\| e^{i\theta_{k_n}} = f_k(\mathbf{x}_k, n) = R_k(\mathbf{x}_k, n) e^{i\Theta_k(\mathbf{x}_k, n)} \quad (13)$$

- For the second case in (Pg. 12), the transformation **without position information** for the query and key, of the token embedding in any position $m, n \in \{1, 2, \dots, L\}$, referred to as the initial condition in (Su et al. 2023), are expressed as follows

$$\mathbf{q} = \|\mathbf{q}\| e^{i\theta_q} = f_q(\mathbf{x}_q, 0) = R_q(\mathbf{x}_q, 0) e^{i\Theta_q(\mathbf{x}_q, 0)} \quad (14)$$

$$\mathbf{k} = \|\mathbf{k}\| e^{i\theta_k} = f_k(\mathbf{x}_k, 0) = R_k(\mathbf{x}_k, 0) e^{i\Theta_k(\mathbf{x}_k, 0)} \quad (15)$$

Regarding the change of notation on token embedding

- $f_q(\mathbf{x}_m, m)$ and $f_k(\mathbf{x}_n, n)$ have been changed to $f_q(\mathbf{x}_q, m)$ and $f_k(\mathbf{x}_k, n)$. Although the purpose of this change in (Su et al. 2023) is unclear, its change here indicates that, with or without position encoding, the same function is used for the transformation of token embedding.
- For the transformation without positional encoding, m and n are set to 0, out of the range of the position in a sequence. If formulated as $f_q(\mathbf{x}_m, m)$, it would become $f_q(\mathbf{x}_0, 0)$, which can not express the transformation at any position $m \in \{1, 2, \dots, L\}$. This applies to $f_k(\mathbf{x}_k, n)$ as well.

Derivation of RoPE in $\mathbb{R}^2$

- Substituting (12,13) into the LHS of (11), which merely changes the formulation of inner product in $\mathbb{R}^2$ to that in $\mathbb{C}$,

$$\begin{aligned}\mathbf{q}_m^\top \mathbf{k}_n &= \operatorname{Re} \left\{ R_q(\mathbf{x}_q, m) R_k(\mathbf{x}_k, n) e^{i[\Theta_k(\mathbf{x}_k, n) - \Theta_q(\mathbf{x}_q, m)]} \right\} \\ &= R_q(\mathbf{x}_q, m) R_k(\mathbf{x}_k, n) \cos(\Theta_k(\mathbf{x}_k, n) - \Theta_q(\mathbf{x}_q, m))\end{aligned}\quad (16)$$

- $g(\mathbf{x}_m, \mathbf{x}_n, n - m)$ is the real part of a complexified vector originally in $\mathbb{R}^2$, because it is used to represent the inner product of two complexified vectors originally in $\mathbb{R}^2$

$$\begin{aligned}g(\mathbf{x}_m, \mathbf{x}_n, n - m) &= \operatorname{Re} \left\{ R_g(\mathbf{x}_q, \mathbf{x}_k, m - n) e^{i[\Theta_g(\mathbf{x}_k, \mathbf{x}_q, m - n)]} \right\} \\ &= R_g(\mathbf{x}_q, \mathbf{x}_k, m - n) \cos(\Theta_g(\mathbf{x}_k, \mathbf{x}_q, m - n))\end{aligned}\quad (17)$$

- Based on (16) and (17), the **condition** in (11) is equivalent to

$$R_q(\mathbf{x}_q, m) R_k(\mathbf{x}_k, n) = R_g(\mathbf{x}_q, \mathbf{x}_k, m - n) \quad (18)$$

$$\Theta_k(\mathbf{x}_k, n) - \Theta_q(\mathbf{x}_q, m) = \Theta_g(\mathbf{x}_k, \mathbf{x}_q, m - n) \quad (19)$$

Derivation of RoPE in $\mathbb{R}^2$

- It is worth noting that the $= g(\mathbf{x}_q, \mathbf{x}_k, m - n)$ in (11) and $= R_g(\mathbf{x}_q, \mathbf{x}_k, m - n)$ in (18) declare the same kind of condition, which is that the LHS's of both depend only on three factors, i.e. $\mathbf{x}_q$, $\mathbf{x}_k$, $m - n$, rather than being equal to any specific parametric form represented as their RHS's.
- This, however, is still effective in guiding the search for the parametric forms of $f_q(\mathbf{x}_q, m)$ and $f_k(\mathbf{x}_k, n)$, even though they may not be the unique or optimal form for query and key, respectively, as will be explained in what follows.

Derivation of RoPE in $\mathbb{R}^2$

- So far, the **goal** of the derivation is to find the parametric forms of $R_q(\cdot), R_k(\cdot), \Theta_q(\cdot), \Theta_k(\cdot)$, which satisfy (18) and (19)
- First, the special cases of $n = m$, where $m \in \{1, 2, \dots, L\} \cup \{0\}$, are considered to find out if it is possible to construct the parametric forms of $R_q(\cdot)$ and $R_k(\cdot)$ for at least these special cases.
- If achieved, the forms will be verified to determine whether they are useful for $n \neq m$. If also achieved, the problems associated with the $R_q(\cdot), R_k(\cdot)$ are solved.
- Then, the same method is applied to $\Theta_q(\cdot), \Theta_k(\cdot)$.

Regarding $m \in \{1, 2, \dots, L\} \cup \{0\}$

- When $m \in \{1, 2, \dots, L\}$, $f_t(\mathbf{x}_t, m)$, where $t \in \{Q, K\}$, is the position encoded transformation of a token embedding. Please refer to equation (20) in (Su et al. 2023) for this interpretation.
- When $m = 0$, $f_t(\mathbf{x}_t, 0)$ is the transformation of a token embedding without position information at any position. Please refer to equation (22) in (Su et al. 2023) for this interpretation.

Derivation of RoPE in $\mathbb{R}^2$

- According to (Pg. 12), the special cases of $n = m$, where $m \in \{1, 2, \dots, L\} \cup \{0\}$, can be further categorized into two subsets
- **The subset of $n = m \neq 0$:** The cases in which the attention weight is computed between the position encoded embedding of $\mathbf{x}_q$ and $\mathbf{x}_k$ in the same position. To satisfy (18),

$$R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, m) = R_g(\mathbf{x}_q, \mathbf{x}_k, 0). \quad (20)$$

- **The subset of $n = m = 0$:** The cases in which the attention weight is computed between the transformation without position information of $\mathbf{x}_q$ and $\mathbf{x}_k$ in any position. To satisfy (18),

$$R_q(\mathbf{x}_q, 0)R_k(\mathbf{x}_k, 0) = R_g(\mathbf{x}_q, \mathbf{x}_k, 0), \quad (21)$$

Further interpretation of (20)

- $\mathbf{x}_q$ and $\mathbf{x}_k$ represent the token embedding at any position. For the same position, $\mathbf{x}_q = \mathbf{x}_k$.
- The 0 on the RHS refers to the same position, not the transformation of a token embedding without position information.
- It should be read as "the product $R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, m)$ of the amplitude parts of $\mathbf{q}_m$, i.e. $f_q(\mathbf{x}_q, m)$, and $\mathbf{k}_m$, i.e. $f_k(\mathbf{x}_k, m)$, at the same position m should depends only on the difference between the position, which is 0."
- In this equation, no transformation without position information is involved yet. In (21), it is.

Derivation of RoPE in $\mathbb{R}^2$

- (20) and (21) can be provided with the $\mathbf{x}_q, \mathbf{x}_k$ in their own respective subsets. In other words, the $\mathbf{x}_q, \mathbf{x}_k$ given as the input of (20) and the $\mathbf{x}_q, \mathbf{x}_k$ given as the input of (21) can be different. In this sense, they are not related.
- But if $\mathbf{x}_q, \mathbf{x}_k$ in a particular position m , which are the same token embedding at m , are given as input to the LHS's of both (20) and (21), ~~their LHS's will be equal because their RHS's are of the same parametric form, although unknown yet, which depends on the same set of three inputs.~~ (wrong!, it should be) their LHS's will depend on the same three factors, which are $\mathbf{x}_q, \mathbf{k}_k$, and 0. In this sense, (20) and (21) are related.
 - The LSH's of (20) and (21), however, are related ONLY because they depend on the same set of factors, NOT because they depend on them in the same way, i.e. not because they have the same parametric form.

Derivation of RoPE in $\mathbb{R}^2$

- Up to this step, the **goal** expressed in (20) and (21) is to construct the parametric forms of $R_q(\mathbf{x}_q, m)$ and $R_k(\mathbf{x}_k, n)$ such that, given $\mathbf{x}_q$ and $\mathbf{x}_k$ in the same position $m \in \{1, 2, \dots, L\}$, their inner product based on their position encoded transformation ~~is the same~~ (wrong! it should be) should depend on the same set of three factors, i.e. $\mathbf{x}_q$, $\mathbf{k}_k$, and 0, as the inner product based on the transformation without position information does. That is, given any $m \in \{1, 2, \dots, L\}$, the two LHS's ~~are equal~~ (wrong, it should be) have the same dependency in terms of the input.
- Nonetheless, it is still acceptable to construct $R_t(\mathbf{x}_t, m)$, $t \in \{Q, K\}$ such that the LHS's of (20) and (21) have the same parametric form for an $m \in \{1, 2, \dots, L\}$, as this will still satisfy the condition of sharing the common dependency. When m is changed, both LHS's will be changed to another value, but still the equal.

- As stated before based on (20) and (21), the goal of constructing $R_q(\mathbf{x}_q, m)$ and $R_k(\mathbf{x}_k, n)$ in the current immediate step is to find out certain parametric forms of them, which satisfy

$$R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, m) = R_q(\mathbf{x}_q, 0)R_k(\mathbf{x}_k, 0), \quad (22)$$

i.e. using a more specific condition that is the equal parametric form as a substitute for the less specific condition that is the shared common dependency in terms of the input.

Derivation of RoPE in $\mathbb{R}^2$

- (22) does not mean that the parametric forms on both sides are found and equal. **The way to understand it is that, whatever parametric form the RHS takes, the form of the LHS should be the same. In other words, all the derivation since (11) just reveals more self-consistency at a finer granularity, i.e. from the $\langle f_q(\mathbf{x}_q, m), f_q(\mathbf{x}_k, n) \rangle$ to $\langle R_q(\mathbf{x}_q, m), R_k(\mathbf{x}_k, n) \rangle$, and from $\forall m, n \in \{1, 2, \dots, L\} \cup \{0\}$ to $m = n = 0$ and $m = n \neq 0$, which guides the search for the ultimate $f_t(\mathbf{x}_t, m)$. In this particular intermediate step, it leads the search towards the form of $R_t(\mathbf{x}_t, m)$ first, $t \in \{Q, K\}$.**
- In addition, (22) only declares that the product on the LHS is equal to that on the RHS, not that some term on the LHS is equal to any term on the RHS.
- Although there may be multiple parametric forms of $R_t(\mathbf{x}_t, m)$, $t \in \{Q, K\}$ satisfying (22), of which the structure, if known, can be used to guide the search for the optimal form of each, it is unnecessary to find all of them.

Derivation of RoPE in $\mathbb{R}^2$

- Therefore, an acceptable construction based on an even further specific condition is,

$$R_t(\mathbf{x}_t, m) = R_t(\mathbf{x}_t, 0), \quad (23)$$

for $t \in \{Q, K\}$. Based on (14) and (15),

$$R_q(\mathbf{x}_q, m) = R_q(\mathbf{x}_q, 0) = \|\mathbf{q}\| \quad (24)$$

$$R_k(\mathbf{x}_k, n) = R_k(\mathbf{x}_k, 0) = \|\mathbf{k}\| \quad (25)$$

- (24, 25) directly result in

$$R_q(\mathbf{x}_q, m)R_k(\mathbf{x}_k, n) = \|\mathbf{q}\| \cdot \|\mathbf{k}\|, \quad (26)$$

satisfying the condition in (18).

- This is exactly the construction formed in (Eq. 26a) in (Su et al. 2023).

Derivation of RoPE in $\mathbb{R}^2$

To be continued ...

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